FIITJEE

ALL INDIA TEST SERIES

FULL TEST – V

JEE (Main)-2020

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 75 questions.
- Each subject (PCM) has 25 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only three sections: **Section-A**, **Section-B and Section-C**.

  Section-A (01 – 20, 26 – 45, 51 – 70) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 22, 46 – 47, 71 – 72) *contains 6 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

  Section-C (23 – 25, 48 – 50, 73 – 75) *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

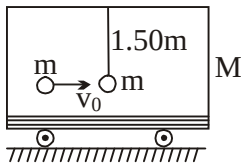
Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A ball of mass m = 1kg is hung vertically by a thread of length ℓ = 1.50 m. Upper end of the thread is attached to the ceiling of a trolley of mass M = 4kg. Initially, trolley is stationary and it is free to move along horizontal rails without friction. A shell of mass m = 1kg, moving horizontally with velocity v_0 = 6 ms^{-1}, collides with the ball and gets stuck with it. As a result, thread starts to deflect towards right. Calculate its maximum deflection with the vertical. (g = 10 ms^{-2})



(A) $37°$ (B) $53°$
(C) $30°$ (D) $60°$

2. The torque ($\vec{\tau} = \vec{r} \times \vec{F}$) of a force **F** = $-3\hat{i} + \hat{j} + 5\hat{k}$ acting at the point **r** = $7\hat{i} + 3\hat{j} + \hat{k}$ is -

(A) $14\hat{i} - 38\hat{j} + 16\hat{k}$ (B) $4\hat{i} + 4\hat{j} + 6\hat{k}$
(C) $-21\hat{i} + 4\hat{j} + 4\hat{k}$ (D) $-14\hat{i} + 34\hat{j} - 16\hat{k}$

3. In a two slit experiment with monochromatic light, fringes are obtained on a screen placed at some distance from the plane of slits. If the screen is moved by 5×10^{-2} m towards the slits, the change in fringe width is 3×10^{-5} cm. If the distance between the slits is 10^{-3} m, calculate the wavelength of the light used.

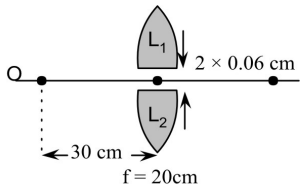
(A) 4500 Å (B) 3000 Å
(C) 9000 Å (D) 6000 Å

4. A certain physical quantity is calculated from the formula $\frac{\pi}{3}(a^2 - b^2)h$, where h, a and b are all length. The quantity being calculated is-

(A) Velocity (B) Length
(C) Area (D) Volume

Space for Rough work

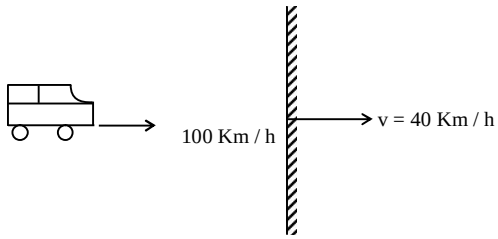
5. A point object O is placed at a distance of 0.3 m from a convex lens (focal length 0.2 m) cut into two halves each to which is displacement by 0.005 m as shown in fig. Find the position of the image. find the distance between the images.



(A) 0.003 m (B) 0.002 m
(C) 0.004 m (D) 0.001 m

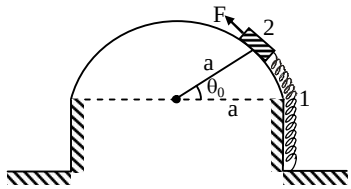
6. A truck starts from rest with an acceleration of 1.5 metre/sec^2 while a car 150 metre behind starts from rest with an acceleration of 2 metre/sec^2. How long will it take before both the truck and car to be side by side, and how much distance is travelled by each?
(A) 2.45 sec (B) 5 sec
(C) 24.5 sec (D) 5.3 sec

7. The back side of the truck which is moving with 40 km/h has shining surface. The surface can be treated as a plane mirror. A motorist trying to overtake the truck from behind is moving with speed of 100 km/h. Find the velocity of the image of motorist observed to himself and to a person standing on the roadside.



(A) 10 kmph (B) 30 kmph
(C) 20 kmph (D) 5 kmph

8. In the position 1 (figure) the spring of constant K is undeformed. Find the work done by the force $\vec{F}$ (which is always directed along the tangent to the smooth hemi spherical surface) on the small block of mass m to shift it from the position 1 to position 2 slowly.

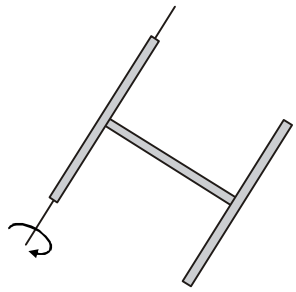


(A) mg R sin $\theta_0 + \frac{KR^2\theta_0^2}{2}$ (B) mg R sin θ_0
(C) mg R sin $\theta_0 + \frac{KR^2\theta_0^2}{4}$ (D) Zero

9. The ionization potential of Li^{++} ion is 122.4 V. Find the value of its first excitation potential.
(A) 91.8v (B) 45.9 V
(C) 122.4V (D) 28.4 V

Space for Rough work

10. A rigid body is made of three identical thin rods, each of length L fastened together in the form of the letter H. The body is free to rotate about a horizontal axis that runs along the length of one of the legs of the H. The body is allowed to fall from rest from a position in which the plane of the H is horizontal. What is the angular speed of the body when the plane of H is vertical $\frac{3}{2}\sqrt{\frac{g}{L}}$



(A) $\frac{5}{2}\sqrt{\frac{g}{L}}$ (B) $\frac{3}{2}\sqrt{\frac{g}{L}}$

(C) $\frac{3}{4}\sqrt{\frac{g}{L}}$ (D) $\sqrt{\frac{g}{L}}$

11. A radioactive sample has 6.0×10^{18} active nuclei at a certain instant. How many of these nuclei will still be in the same active state after two half-lives?

(A) 3×10^{18} (B) 1.5×10^{18}

(C) 0.75×10^{18} (D) 6×10^{18}

12. One solid sphere, having mass 1kg & diameter 0.3m is suspended from a wire. If the twisting couple per unit twist for the wire is 6×10^{-3} N-m/radian, then the time period of small oscillations will be:

(A) 0.7 sec (B) 7.7 sec

(C) 77 sec (D) 777 sec

13. Find the constant b in Moseley's equation $\sqrt{\nu} = a(Z - b)$ from the following data.

Element	Z	Wavelength of K_a X-ray
Mo	42	71 pm
Co	27	178.5 pm

(A) 1.43 (B) 1.11

(C) 1 (D) 1.37

14. A 70 cm long sonometer wire is in unison with a tuning fork. If the length of the wire is decreased by 1.0 cm, it produces 4 beats per sec. with the same tuning fork. Find the frequency of the tuning fork -

(A) 276 sec^{-1} (B) 272 sec^{-1}

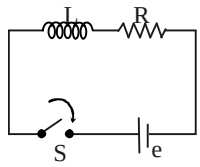
(C) 269 sec^{-1} (D) 280 sec^{-1}

Space for Rough work

15. Mr. Sanjay wants to calculate the current and power dissipated in an LCR series circuit. He connected 100W resistance to an AC source of 200 V and angular frequency 300 radian/sec. When he removed only the capacitance, the current was found to be lagging behind the voltage by 60°. While on removing the inductance he found, the current leading the voltage by 60°. Find the value of current and the power dissipated obtained by him
(A) 200 Watt (B) 100 Watt
(C) 400 Watt (D) 300 Watt

16. The length of a wire between the two ends of a sonometer is 105 cm. Where should the two bridges be placed so that the fundamental frequencies of the three segments are in the ratio of 1:3:15.
(A) 75 cm, 25 cm, (B) 25 cm, 75 cm,
(C) 75 cm, 100 cm, (D) None of these

17. In an LR circuit as shown in fig. when the switch is closed how much time will it take the current to grow a value $\frac{\eta\varepsilon}{R}$ (where η < 1)



(A) (L/R) In $\left|\frac{1}{\eta}\right|$ (B) (L/R) In $\left|\frac{1}{1-\eta}\right|$
(C) (2L/R) In $\left|\frac{1}{\eta}\right|$ (D) None of these

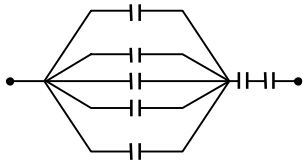
18. An infinite number of charges each equal to q are placed along the x-axis at x = 1, x = 2, x = 4, x = 8 ..... and so on. Find the electric field at the point x = 0 due to this set of charges.
(A) $\frac{1}{4\pi\varepsilon_0}\left[\frac{2}{3}q\right]$ (B) $\frac{1}{4\pi\varepsilon_0}\left[\frac{8}{3}q\right]$
(C) $\frac{1}{4\pi\varepsilon_0}\left[\frac{4}{3}q\right]$ (D) $\frac{1}{4\pi\varepsilon_0}\left[\frac{5}{3}q\right]$

19. A large plane charge sheet having surface charge density σ = 2.0 × 10^{-6} C/m^2 lies in the X-Y plane. Find the flux of the electric field through a circular area of radius 1 cm lying completely in the region where x, y, z are all positive and with its normal making an angle of 60° with the Z-axis.
(A) 15 N-m^2/C (B) 12.5 N-m^2/C
(C) 17.5 N-m^2/C (D) None of these

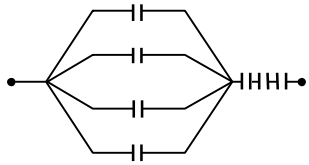
Space for Rough work

20. Seven capacitors each of capacitance 2 mF are to be connected in a configuration to obtain an effective capacitance of (10/11)mF. Which of the combination(s) shown in fig. will achieve the desired result?

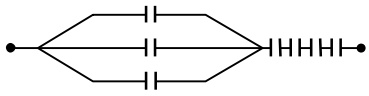
(A)



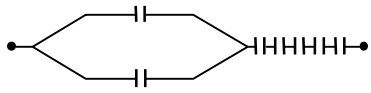
(B)



(C)



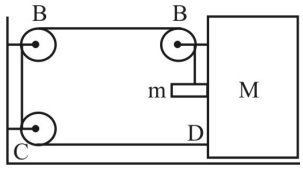
(D)



SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

21. In figure the acceleration of m is $\frac{2\sqrt{x}mg}{M+5m}$, assuming that there is no friction between m and M, and all other surfaces are smooth and pulleys light. Find the value of x.



22. A non-conducting thin spherical shell of radius R has uniform surface charge density σ. The shell rotates about a diameter with constant angular velocity ω. The magnetic induction B at the centre of the shell is $\frac{2}{x}\mu_0\ \sigma\omega$ R. Find the value of x.

Space for Rough work

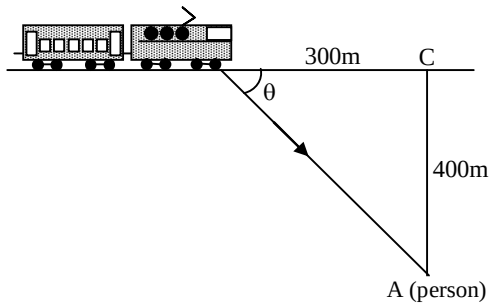
SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

23. A gun kept on a straight horizontal road is used to hit a car travelling along the same road away from the gun with a uniform speed of 72 km h^{-1}. The car is at a distance of 500 m from the gun, when the gun is fired at an angle of 45° with the horizontal. Find the speed of projection of the shell from the gun. (in m/s^{-1})

24. An ideal monatomic gas is confined in a cylinder by a spring loaded piston of cross-section 8×10^{-3} m^2. Initially the gas is at 300K and occupies a volume of $2.4 \times 10^{-3} m^3$ and the spring is in its relaxed position. The gas is heated by a small electric heater until the piston moves out slowly by 0.1m. Calculate the heat supplied (in J) by the heater. The force constant of the spring is 8000 N/m and atmospheric pressure 1×10^5 N/m^2.

25. A train approaching a railway crossing at a speed of 120 km/h sounds a short whistle at frequency 640 Hz when it is 300m away from the crossing. The speed of sound in air is 340 m/s. What will be the frequency heard by a person standing on a road perpendicular to the track through the crossing at a distance of 400 m from the crossing. (in Hz)



Space for Rough work

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

26. Most soluble hydroxide in benzene is:
(A) $LiOH$ (B) $NaOH$
(C) KOH (D) $RbOH$

27. Referring to figure-1, identify the three spots obtained by electrophoresis at $p^H = 7$ of a mixture of the proteins pepsin $\left(p^I = 1.1\right)$ haemoglobin $\left(p^I = 6.8\right)$ and Lysozyme $\left(p^I = 11.0\right)$:

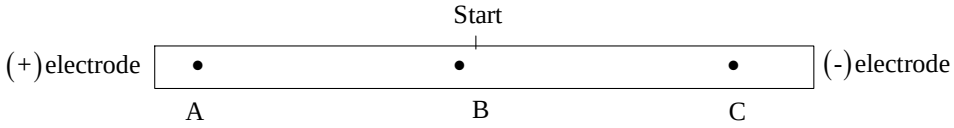


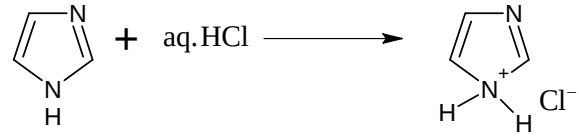
Fig-1

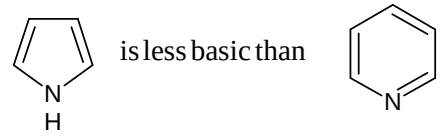
Which of the following options are correct about position of various proteins?
(A) Lysozyme will be present at spot A
(B) pepsin will be present at spot B
(C) pepsin will be present at spot A
(D) haemoglobin will be present at spot C

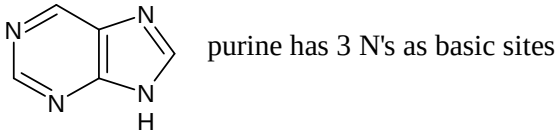
28. Silver sulphide dissolves in sodium cyanide solution to form the complex?
(A) $Na_2\left[Ag\left(CN\right)_4\right]$ (B) $Na\left[Ag\left(CN\right)_2\right]$
(C) $Na_3\left[Ag\left(CN\right)_4\right]$ (D) $Na_3\left[Ag\left(CN\right)\right]$

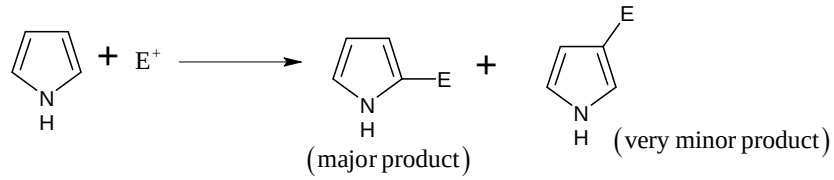
Space for Rough work

29. Which of the following is incorrect statement/reaction?

(A) 

(B) 

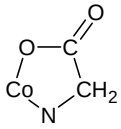
(C) 

(D) 

30. Which of the following statement is wrong about complex anion $\left[Co(EDTA)\right]^-$?

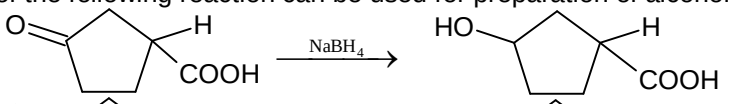
(A) oxidation state of Co is $+3$

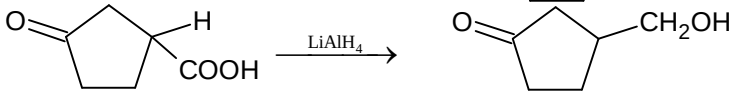
(B) compound is optically active

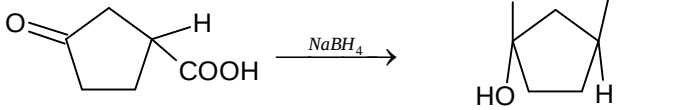
(C) number of  rings is 4

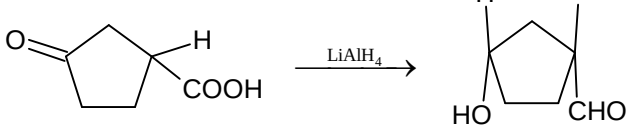
(D) total number of five member rings is 6

31. Which of the following reaction can be used for preparation of alcohol from carbonyl compound

(A) 

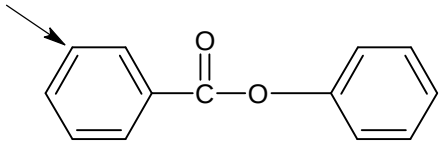
(B) 

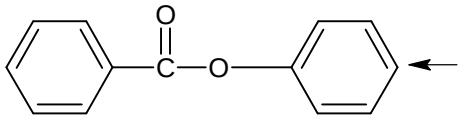
(C) 

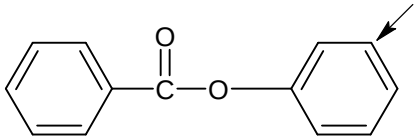
(D) 

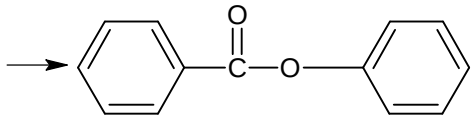
Space for Rough work

32. When phenyl benzoate ($PhOCOPh$) undergoes electrophilic substitution, which of the following indicates the site for attack of electrophile as major product?

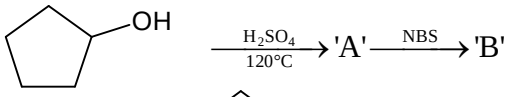
(A) 

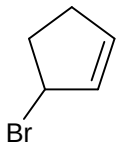
(B) 

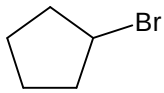
(C) 

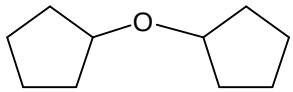
(D) 

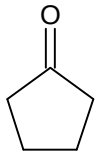
33. Which of the following statement about diffusion of gas is true?
(A) Rate of diffusion of a gas will increase if an inert gas is added to gaseous mixture and allowed to diffuse through a pin hole
(B) Rate of effusion of O_2 is $\sqrt{2}$ times that of SO_2, for every composition of O_2 and SO_2 mixture
(C) Rate of effusion decrease on increasing the temperature
(D) All are true.

34. 

(A) 'B' is 

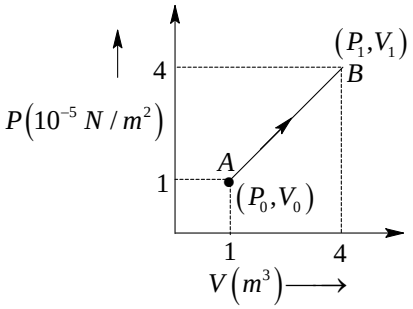
(B) 'B' is 

(C) 'A' is 

(D) 'A' is 

Space for Rough work

35. The work done in changing state from A to B in following system is:



(A) $-7.5\times10^{-5}\,J$ (B) $-7.5\times10^{5}\,J$

(C) $+7.5\times10^{-5}\,J$ (D) $+7.5\times10^{5}\,J$

36. $P_4 + HNO_3 \longrightarrow H_3PO_4 + NO_2\uparrow + H_2O$

Product mixture is titrated with $0.1M$ $NaOH$, indicator shows colour change at second end point and $100ml$ of $NaOH$ is used for that. How much volume of $1M$ HNO_3 is used is above reaction?

(A) $20ml$ (B) $10ml$

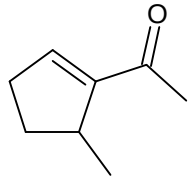
(C) $25ml$ (D) $\frac{50}{3}ml$

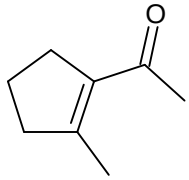
37. When Lithium di-(3-pentyl) cuprate reacts with ethyl bromide produces:

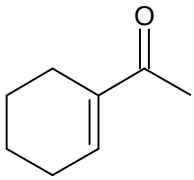
(A) heptane (B) triethyl methane

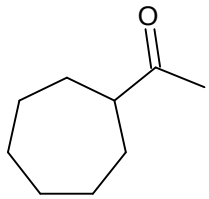
(C) 2-ethylpentane (D) 3-methyl hexane

38. 2, 7-Octanedione on treatment with aq. KOH and 100°C produces?

(A) 

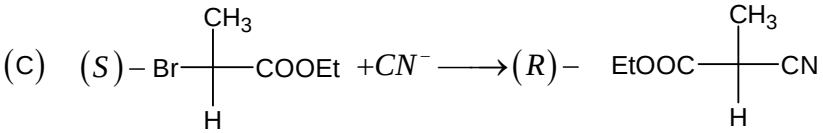
(B) 

(C) 

(D) 

Space for Rough work

39. Which of the following salts are soluble in NH_4OH?
(A) $CuSO_4$ (B) $AlCl_3$
(C) $FeCl_3$ (D) all of the above

40. Which of the following substitution reaction $\left(SN^2\right)$ does not give product as shown is reactions?
(A) $(R)-CH_3CH(Br)CH_2CH_3+MeO^- \longrightarrow (S)-CH_3CH(OMe)CH_2CH_3$
(B) $(S)-CH_3CH(Br)CH_2CH_3+MeO^- \longrightarrow (R)-CH_3CH(OMe)CH_2CH_3$
(C) 
(D) cis-4-iodoethylcyclohexane+OH^- $\longrightarrow$ trans-4-ethylcyelohexanol

41. The aqueous solution of which salt is coloured
(A) $Zn(NO_3)_2$ (B) $LiNO_3$
(C) $Co(NO_3)_2$ (D) Potash alum

42. Potassium crystallizes in BCC lattice with a unit cell length $a=5.0\text{Å}$. What is the distance between next nearest neighbours?
(A) 5.0Å (B) 4.33Å
(C) 3.5Å (D) 7.07Å

43. Crystal field splitting energy (Δ_0) of complexes $\left[CrCl_6\right]^{3-}$, $\left[Cr(CN)_6\right]^{3-}$ and $\left[Cr(NH_3)_6\right]^{3+}$ are x, y and z cm^{-1} respectively. Then correct order Δ_0 values is:
(A) $x<y<z$ (B) $x<z<y$
(C) $x>y>z$ (D) $x>z>y$

44. Optical isomerism is not shown by the complex:
(A) $\left[Cr(ox)_3\right]^{3-}$ (B) $Cis\left[Co(en)_2Cl_2\right]^+$
(C) $Trans\left[Co(en)_2Cl_2\right]^+$ (D) $\left[Cr(en)_3\right]^{3+}$

Space for Rough work

45. What concentration of Ag^+ is solution will just fail to give a precipitate of Ag_2CrO_4 with a solution in which $\left[CrO_4^{2-}\right] = 3\times10^{-4}M$?

$\left(Given\ K_{sp}\left(Ag_2CrO_4\right) = 1.2\times10^{-11}\right)$

(A) $10^{-3}M$ (B) $2\times10^{-1}M$

(C) $2\times10^{-3}M$ (D) $2\times10^{-4}M$

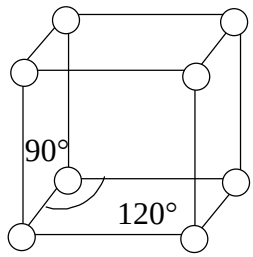
SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

46. How many statement(s) are correct?

(a) if the critical micelle concentration (CMC) of a soap is $10^{-3}M$, then $10^{-4}M$ solution of this soap will have colloidal nature.
(b) charge on the colloidal particles may be determined with the help of dialysis
(c) peptization is the process of conversion of a colloidal sol into precipitate by adding some electrolyte
(d) when some oil-soluble dye is added in the water in oil emulsion, the back ground becomes coloured
(e) For the coagulation of a particular colloidal solution, the order of coagulation power of some electrolytes is in the order: $Na_3PO_4 > BaSO_4 > AlCl_3$, in electro-osmosis, the dispersion medium move towards the anode
(f) diffused layer around colloidal particles in the electric double layer mostly contains negatively charged particles.(colloid is of 'e' part)
(g) it may be basic dye such as methylene blue (colloid is of 'e' part)
(h) it may be metal sol (colloid is of 'e' part)

47. Ice crystallizes is an hexagonal lattice. At the low temperatures at which the structure was determined, the lattice constants were $a = 4.53\text{Å}$ and $b = 7.41\text{Å}$ as shown in the figure. How many H molecules are contained in a unit cell? (Given that density of ice $= 0.92\ g/cm^3$)?



Space for Rough work

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

48. If value of spin quantum number(s) $= -1/2, 0, +1/2$ then calculate number of groups in the new form of periodic table if all other rules of electronic configuration are remain same.

49. Calculate value of $\frac{X+Y+Z}{10}$, here X is $O-N-O$ bond angle in NO_3^- Y is $O-N-O$ bond angle in NO_2^+ and Z is $F-Xe-F$ adjacent bond angle in XeF_4.

50. Calculate $x+y+z$ for H_3PO_3 acid, where x is number of lone pairs, y in number of σ bonds and z in number of π bonds.

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

51. $\int \frac{3x^2+4x-1}{\left(x^2+1\right)^2\sqrt{x+1}}dx$ is equal to:

(A) $\frac{\sqrt{x+1}}{x^2+1}+c$ (B) $-\frac{2\sqrt{x+1}}{x^2+1}+c$

(C) $\frac{2}{\left(x^2+1\right)\sqrt{x+1}}$ (D) $\frac{-x}{\left(x^2+1\right)\sqrt{x+1}}+c$

52. The general solution of the differential $y\left(xy-\left(x^2-y^2\right)^2\right)dx=\left\{y^3-x\left(x^2-y^2\right)^2\right\}dy$ is:

(A) $\frac{y}{x}+\frac{1}{x^2-y^2}=c$ (B) $\frac{x}{y}-\frac{1}{2}\ln\left|\frac{x-y}{x+y}\right|=c$

(C) $\frac{2x}{y}+\frac{1}{x^2-y^2}=c$ (D) $\frac{2x}{y}-\frac{1}{x^2-y^2}=c$

53. A line is drawn at an angle θ with positive direction of x-axis. It intersects the parabola $y^2=8x$ at A and B. If AB is a normal chord which subtend a right angle at the vertex of the parabola, then $\tan^2\theta$ equals:

(A) $\frac{1}{2}$ (B) $\frac{3}{4}$

(C) 1 (D) 2

Space for Rough work

54. The equation of a circle whose diameter is the common chord of the circle $x^2+y^2-4x-8y-5=0$ and $x^2+y^2-12x+2y+29=0$ is:

(A) $41x^2+41y^2-396x-38y+781=0$

(B) $2x^2+2y^2-16x-6y+25=0$

(C) $41x^2+41y^2+396x-38y-781=0$

(D) $17x^2+17y^2-300x+154y+901=0$

55. $\int_0^{\frac{11\pi}{3}}\sqrt{1+\cos 2x}\,dx$ is equal to:

(A) $\frac{11\sqrt{3}}{2}$ (B) $8\sqrt{2}-\sqrt{\frac{3}{2}}$

(C) $8\sqrt{2}-\frac{\sqrt{3}}{2}$ (D) $\frac{22}{3}$

56. Consider the vectors $\vec{a}=2\hat{i}+\hat{j}-2\hat{k}$ and $\vec{b}=3\hat{i}-\hat{j}+\hat{k}$. A parallelogram is constructed such that its diagonal are along $2\vec{a}-\vec{b}$ and $\vec{a}-2\vec{b}$. The area of the parallelogram is:

(A) $9\sqrt{5}\ sq.units$ (B) $\frac{9\sqrt{10}}{2}\ sq.units$

(C) $9\sqrt{10}\ sq.units$ (D) $18\sqrt{5}\ sq.units$

57. Let $\frac{\tan\left(\frac{\pi}{4}+\alpha\right)}{5}=\frac{\tan\left(\frac{\pi}{4}+\beta\right)}{3}=\frac{\tan\left(\frac{\pi}{4}+\gamma\right)}{2}$. Then $12\sin^2(\alpha-\beta)+15\sin^2(\beta-\gamma)-7\sin^2(\gamma-\alpha)$ is equal to?

(A) $-\frac{1}{2}$ (B) $\frac{1}{2}$

(C) 1 (D) 0

Space for Rough work

58. The angle of intersection of the curves $y=(1+x)^{\cos x}+\sin x$ and $y=\frac{1}{2}\left(x^2+x+2\right)$ at (h,k) where h and k are integers, is θ, then $\tan\theta$ equals:

(A) $\frac{3}{4}$ (B) $\frac{4}{3}$

(C) 1 (D) 0

59. A plane $2x+3y+6z=12$ intersects the x-axis, y-axis and z-axis at A, B, C respectively. The orthocentre of the triangle ABC is:

(A) $\left(\frac{24}{49},\frac{36}{49},\frac{72}{49}\right)$ (B) $\left(\frac{36}{49},\frac{72}{49},\frac{27}{49}\right)$

(C) $\left(\frac{24}{49},\frac{16}{49},\frac{1}{49}\right)$ (D) $\left(\frac{36}{49},\frac{72}{49},\frac{24}{49}\right)$

60. The domain of the function given by $f(x)=\log_3\left(\log_{\frac{1}{2}}\left(\log_2\left(\log_{\frac{1}{3}}(x)\right)\right)\right)+\frac{x+1}{\sqrt{1-3x-4x^2}}$ is:

(A) $\left(-1,\frac{1}{3}\right)$ (B) $\left(\frac{1}{4},\frac{1}{3}\right)$

(C) $\left(\frac{1}{9},\frac{1}{4}\right)$ (D) $\left(\frac{1}{9},\frac{1}{3}\right)$

61. If $f(x)=\frac{3^{x-3}}{3^{1-x}+3^x}$ for all $x\in Q$, then the value of $f\left(\frac{1}{109}\right)+f\left(\frac{2}{109}\right)+f\left(\frac{3}{109}\right)+....+f\left(\frac{108}{109}\right)$ is:

(A) 54 (B) 55

(C) 108 (D) 2

Space for Rough work

62. The value of $\cot\sum_{n=1}^{\infty}\cot^{-1}\left(\frac{2}{n^2}\sum_{k=1}^{n}k^3\right)$

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$

(C) $\frac{1}{3}$ (D) $\frac{1}{6}$

63. The graph of the function $y=f(x)$ has a unique tangent at $(\pi^a,0)$. If $A=\lim_{x\to\pi^a}\frac{\ln(1+9f(x))-\sin f(x)}{2f(x)}$, then $\sum_{n=1}^{\infty}A^{-n}$ is equal to:

(A) $\frac{1}{3}$ (B) $\frac{1}{2}$

(C) 4 (D) $\frac{1}{4}$

64. Let a_r, $r=1,2,3,4$ represent four positive real numbers other than unity such that $\left(\log_e a_r,\frac{1}{\log_e,a_r}\right)$ is a point on a circle whose abscissa of end of diameter are α and $-\alpha$, then the minimum value of $a_1+a_2+a_3+a_4$ is:

(A) 2 (B) 3

(C) 4 (D) 5

65. The chord of contact of tangents drawn from any point on the circle $x^2+y^2=p^2$ to the circle $x^2+y^2=q^2$ touches the circle $x^2+y^2=r^2$, then $2a-\log_e p, 4a-\log_e q, 6a-\log_e r$ are in (a is real value):

(A) AP (B) GP

(C) HP (D) AGP

Space for Rough work

66. If t_1, t_2 and t_3 are distinct, the points $\left(t_1, 2at_1 + at_1^3\right), \left(t_2, 2at_2 + at_2^3\right)$ and $\left(t_3, 2at_3 + at_3^3\right)$ are collinear, then $\frac{4^{t_1}.6^{t_2}.8^{t_3}}{2^{t_1}.3^{t_2}.4^{t_3}}$ is equal to:

(A) 0 (B) 4

(C) 2 (D) 1

67. The area of the triangle formed by the line $x + y = 2$ and angle bisectors of the pair of straight lines $x^2 - y^2 + 2y = 1$ is a and coordinate of orthocentre of triangle is (b, c), then $4a + b + c$ is equal to:

(A) 0 (B) 4

(C) 2 (D) 3

68. If z is the complex 5th root of unity, then $\frac{z^4}{z^4+1} + \frac{z}{1+z} + \frac{z^2}{z^2+1} + \frac{z^4}{z+z^4}$ is:

(A) 2 (B) 0

(C) 4 (D) 1

69. If a tangent of slope 3 of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is normal to the circle $x^2 + y^2 + 4x + 1 = 0$ then the maximum value of ab is:

(A) 2 (B) 3

(C) 4 (D) 6

70. If $F(x) = \int_0^x f(t).dt$ and $F\left(x^2\right) = x^2(1+x)$, then value of $f(4) = n$. Then unit place of $(985697)^{n^2}$ is:

(A) 3 (B) 7

(C) 1 (D) 9

Space for Rough work

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

71. A nine digit number is formed from $1,2,3,4,5$ such that product of all digits is always 1920. The total number of ways is $393\left({}^{9}P_{r}\right)$, where the value of r is_____

72. The sum $\sum_{i=0}^{5m}\binom{20}{i}\binom{30}{5m-i}$, ${}^{p}C_{q}=\binom{p}{q}=0$ if $p<q$, is maximum, then m equals_____

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

73. Let $f(x)$ be a function such that $f(x)+f(x+2)=\sqrt{3}f(x+1)$ for all $x\in R$, then f is periodic with period T. Then value of T is_____

74. If the sum of first n terms of the A. P. $85,90,95,....$ is equal to the sum of first 3n terms of the A. P. $9,11,13,...,$ then $n+10$ equals_____

75. Let f be a continuous and differentiable function in (a,b), if $f(x).f'(x)\geq x\sqrt{1-\left(f(x)\right)^{4}}$ and $\lim_{x\to a^{+}}\left(f(x)\right)^{2}=1$ and $\lim_{x\to b^{-}}\left(f(x)\right)^{2}=\frac{1}{2}$. Then the minimum value of $\left[a^{2}-b^{2}\right]+80$ is$\left([.]\to\text{represent greatest integer function}\right)$:

Space for Rough work